

Poisson Distribution Crash Course

1 Definition & Intuition

The **Poisson distribution** models the **number of events** occurring in a fixed interval (time, space, volume, etc.) when events happen:

- Independently of one another
- At a constant average rate

Key Insight

Poisson answers: “How many events will happen in this interval?”

2 PMF (Probability Mass Function)

For $X \sim \text{Poisson}(\lambda)$:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k = 0, 1, 2, 3, \dots$$

where:

- λ = average number of events in the interval
- k = actual number of events we’re calculating probability for
- $e \approx 2.71828$ (Euler’s number)

3 Mean & Variance

$$\begin{aligned}\mathbb{E}[X] &= \lambda \\ \text{Var}(X) &= \lambda \\ \text{SD}(X) &= \sqrt{\lambda}\end{aligned}$$

Special Property: Mean = Variance = λ

4 When to Use Poisson

4.1 Classic Examples

- Number of emails received per hour
- Number of customers arriving at a store per day
- Number of phone calls to a call center per minute
- Number of defects in a manufactured product
- Number of earthquakes in a region per year
- Number of mutations in a DNA sequence

4.2 Requirements

1. **Independence:** Events occur independently
2. **Constant rate:** Average rate λ is constant throughout the interval
3. **No simultaneity:** Two events cannot occur at exactly the same instant

5 Key Properties

5.1 Additivity

If $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ are independent, then:

$$X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$$

5.2 Approximation to Binomial

When n is large and p is small (roughly $n \geq 20$ and $p \leq 0.05$):

$$\text{Binomial}(n, p) \approx \text{Poisson}(np)$$

5.3 Scaling

If $X \sim \text{Poisson}(\lambda)$ counts events in time t , then events in time ct follow:

$$\text{Poisson}(c\lambda)$$

6 Relationship to Exponential Distribution

Crucial Connection

Poisson counts **how many** events in fixed time

Exponential measures **time between** consecutive events

If events arrive as a Poisson process with rate λ per unit time:

- Number of events in time t : $\sim \text{Poisson}(\lambda t)$
- Time between consecutive events: $\sim \text{Exponential}(\text{mean} = 1/\lambda)$

6.1 Exponential Distribution Recap

For $T \sim \text{Exponential}(\text{mean} = \mu)$:

$$\text{PDF: } f(t) = \frac{1}{\mu} e^{-t/\mu} \quad \text{for } t \geq 0$$

$$\text{CDF: } F(t) = 1 - e^{-t/\mu}$$

$$\text{Survival: } S(t) = e^{-t/\mu}$$

Alternative parameterization with rate $\lambda = 1/\mu$:

$$\text{PDF: } f(t) = \lambda e^{-\lambda t}$$

$$\text{Survival: } S(t) = e^{-\lambda t}$$

7 Worked Examples

7.1 Example 1: Basic Calculation

A call center receives an average of $\lambda = 3$ calls per minute.

Q: What's the probability of exactly 2 calls in the next minute?

Solution:

$$\begin{aligned} P(X = 2) &= \frac{3^2 e^{-3}}{2!} \\ &= \frac{9 \times 0.0498}{2} \\ &= \frac{0.4482}{2} \\ &= 0.224 \end{aligned}$$

7.2 Example 2: Zero Events

With $\lambda = 3$ calls per minute, what's the probability of *no* calls?

Solution:

$$\begin{aligned} P(X = 0) &= \frac{3^0 e^{-3}}{0!} \\ &= e^{-3} \\ &\approx 0.0498 \end{aligned}$$

7.3 Example 3: At Least One Event

What's the probability of at least 1 call?

Solution:

$$\begin{aligned} P(X \geq 1) &= 1 - P(X = 0) \\ &= 1 - e^{-3} \\ &\approx 0.950 \end{aligned}$$

7.4 Example 4: Time Between Events

If calls arrive as Poisson($\lambda = 3$ per minute), what's the average time between calls?

Solution: Time between calls $\sim$ Exponential(mean = $1/\lambda = 1/3$ minute = 20 seconds)

Bonus: What's the probability the next call arrives within 30 seconds (0.5 minutes)?

$$\begin{aligned}P(T \leq 0.5) &= 1 - e^{-\lambda t} \\&= 1 - e^{-3 \times 0.5} \\&= 1 - e^{-1.5} \\&\approx 0.777\end{aligned}$$

7.5 Example 5: Scaling

If $\lambda = 3$ calls per minute, how many calls do we expect in 5 minutes?

Solution: Number of calls in 5 minutes $\sim$ Poisson($5 \times 3 = 15$)

Expected number: $\mathbb{E}[X] = 15$ calls

8 Common Pitfalls

1. **Don't forget $k!$ in denominator** - It's easy to drop when calculating by hand
2. **Rate vs. mean confusion** - In exponential: if Poisson rate is λ , exponential mean is $1/\lambda$
3. **Scaling the rate** - If $\lambda = 2$ per hour, then for 3 hours use $\lambda = 6$
4. **Independence assumption** - Poisson requires events to be independent (e.g., doesn't work well for bus arrivals that run on a schedule)

9 Quick Reference Table

Property	Formula/Value
PMF	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$
Mean	λ
Variance	λ
Standard Deviation	$\sqrt{\lambda}$
$P(X = 0)$	$e^{-\lambda}$
$P(X \geq 1)$	$1 - e^{-\lambda}$
Related Exponential Mean	$1/\lambda$